

ADGA Referral Commission System

Client Requirement → Demo Solution → End-to-End Working Process → Real ADGA Implementation Plan

Demo objective	Prove that ADGA can track a referral journey from B (referrer) to C (customer), calculate commission, and give A (owner) complete visibility.
Current demo rate	10% commission, configurable so the percentage can be changed later.
Demo service	Skillwise Demo Service — ₹10,000 purchase used for end-to-end testing.
Demo result	₹10,000 purchase → 10% commission → ₹1,000 commission recorded and visible to both B and A.

Executive summary

The requested referral concept was converted into a standalone proof-of-concept so the complete business flow could be tested independently before integrating it into the live ADGA website. The demo validates referral-link attribution, customer registration, customer purchase, commission calculation, referrer visibility, and owner/admin visibility.

1. Client Requirement

The requirement discussed with the client was centered on a three-party referral relationship:

- A — Owner / Company: should be able to see what is happening across the referral journey.
- B — Referrer: shares a unique referral link with another person.
- C — Referred Customer: opens the referral link, registers, and may purchase a service.
- The system should track whether C actually did something and, specifically, whether C purchased a service.
- The current demonstration uses a 10% commission rate; the rate should remain configurable because the percentage may change later.
- The initial proof-of-concept uses a demo service so the business logic can be validated without affecting the live ADGA payment flow.

Target business flow

A (Owner) has visibility into the whole process. B (Referrer) gets a unique link. C (Customer) arrives through that link, registers, and purchases. The system attributes the purchase to B and calculates the configured commission.

B — Referrer	→	Unique Referral Link	→	C — Customer
Gets REF001		ref=REF001		Registers + purchases
A — Owner / Admin	←	Tracking + Commission	←	Purchase Event

2. Problem Identified and How It Was Solved

The main problem was not only generating a referral link. The important requirement was to maintain attribution across the whole customer journey: link click → registration → purchase → commission → reporting. A simple static referral URL would not give the owner reliable visibility into the business outcome.

Solution used in the demo: a dedicated referral record and user attribution flow connects the referrer code to the referred customer and then to the purchase. The purchase endpoint calculates commission from the configured percentage and stores the resulting referral activity for reporting.

- Attribution: referral code such as REF001 is carried from B's link into C's registration.
- Tracking: referral clicks and registration/purchase events are recorded.
- Commission: calculated automatically from the configured rate.
- Visibility: B sees their own activity; A sees all referral activity.

3. End-to-End Demo Process

Step 1 — B logs in as Referrer

B logs into the Referrer area using the demo referrer account. The dashboard displays the unique referral link, current clicks, registrations, purchases, and commission.

Step 2 — B shares the referral link

The dashboard generates a link such as <http://localhost:5173/?ref=REF001>. B can copy this link and share it with C.

Step 3 — C opens the referral link

C opens the link in a new/incognito browser session. The demo detects REF001 and confirms that the referral link has been detected.

Step 4 — C registers

C creates a customer account. The referral code is carried into the registration process so the customer is attributed to B.

Step 5 — C logs in

After registration, C logs in through the Customer Login page. The authenticated customer can access the demo service catalogue.

Step 6 — C purchases a service

C selects the Skillwise Demo Service priced at ₹10,000 and clicks Purchase Demo. This simulates the business purchase event.

Step 7 — Commission is generated

The system applies the configured 10% rate: $₹10,000 \times 10\% = ₹1,000$. The purchase and commission are recorded.

Step 8 — B sees the result

The Referrer dashboard updates to show 1 click, 1 registration, 1 purchase, and ₹1,000 commission, along with the customer/service activity.

Step 9 — A checks the complete process

The Owner/Admin dashboard shows 1 click, 1 registration, 1 purchase, ₹10,000 revenue, and ₹1,000 commission. The activity row shows Demo Referrer → Demo Customer → Skillwise Demo Service → ₹10,000 → ₹1,000 → Purchased.

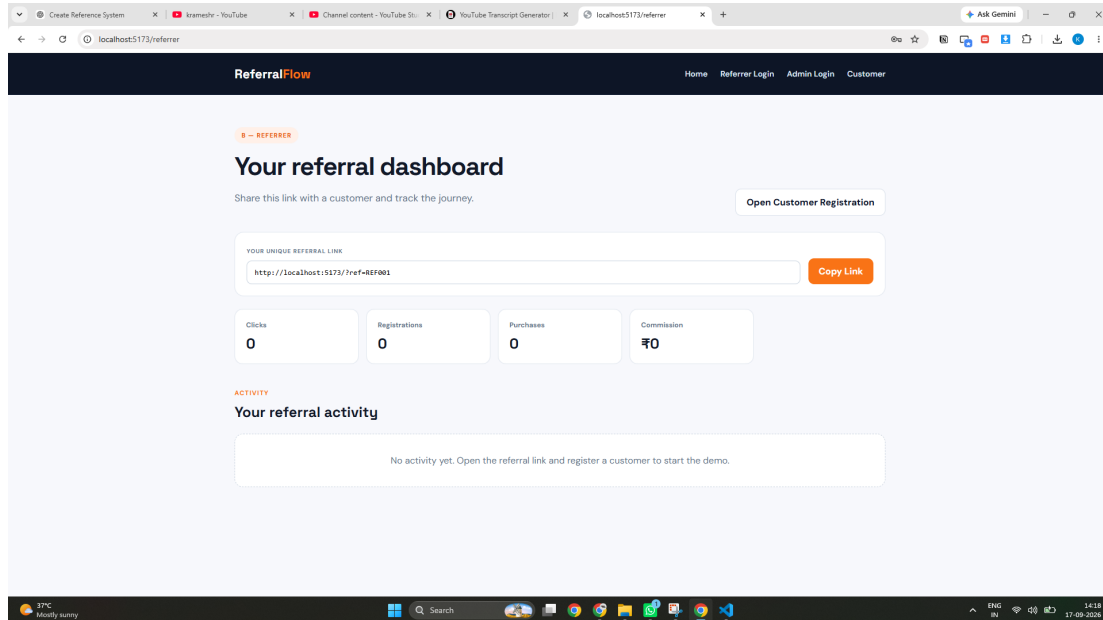
Verified calculation

Purchase	Commission rate	Commission
₹10,000	10%	₹1,000

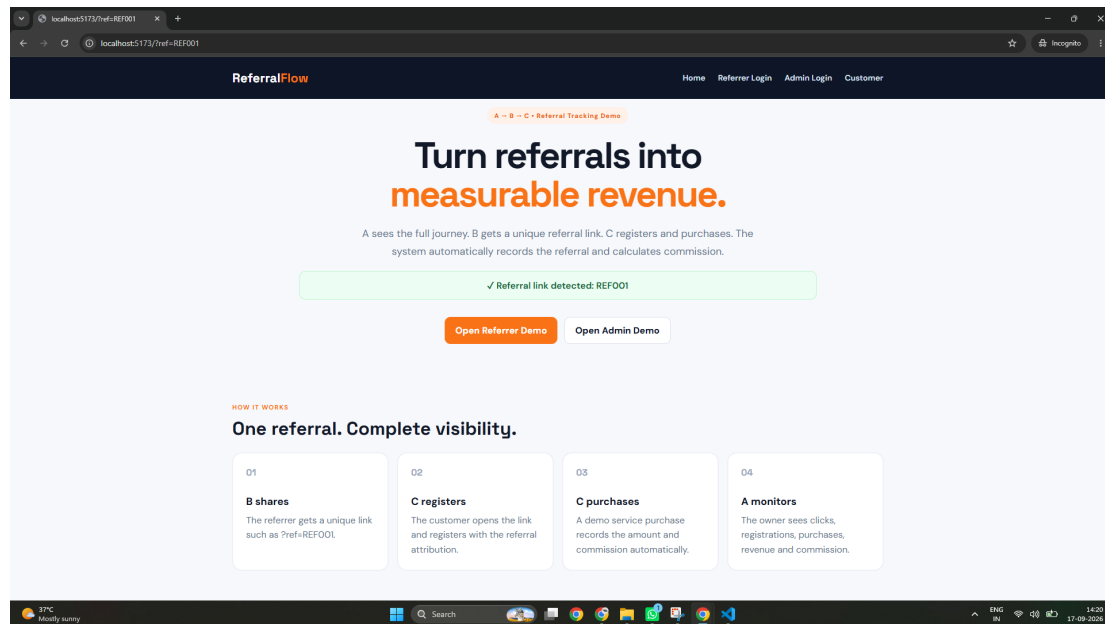
4. Demo Evidence — Working Screens

The following screenshots document the tested flow. They are included as implementation evidence for the proof-of-concept.

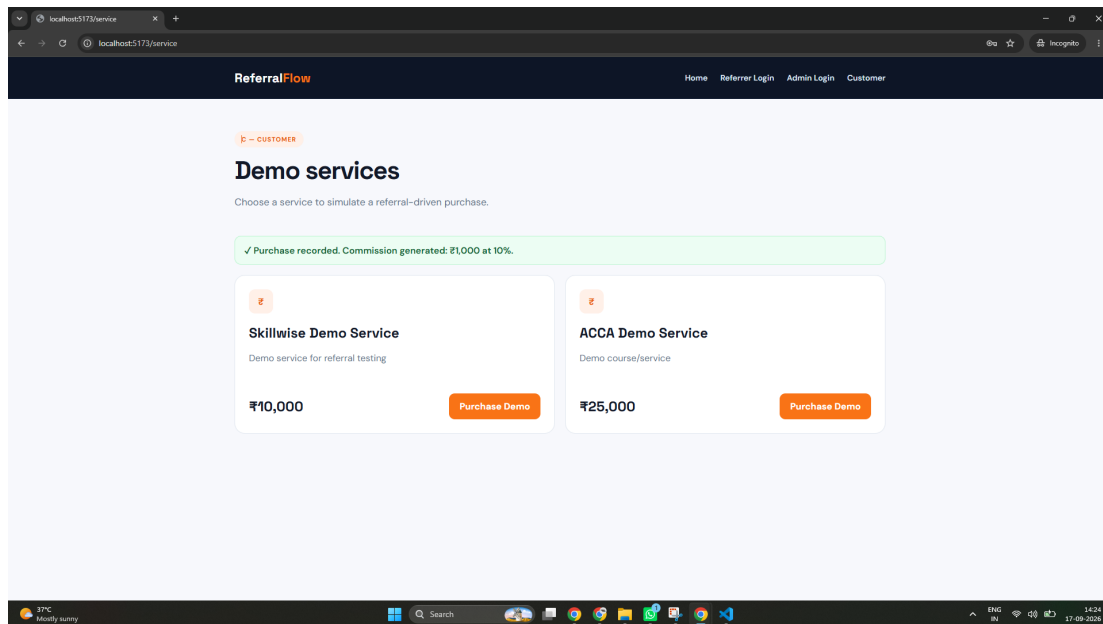
B — Referrer dashboard before activity



C — Referral link detected in a new browser session



C — Purchase recorded and ₹1,000 commission generated

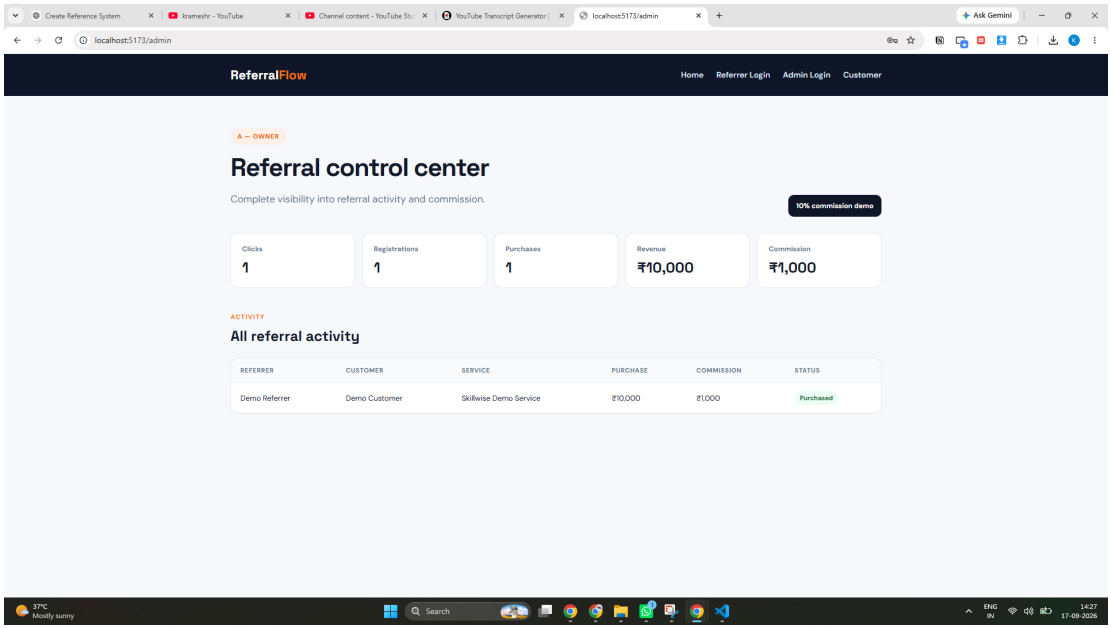


B — Referrer dashboard after purchase

The screenshot shows a web browser window with the URL `localhost:5173/referrer`. The page is titled "ReferralFlow" and has a navigation bar with "Home", "Referrer", and "Logout" links. The main content area is titled "Your referral dashboard" and includes a sub-header "Share this link with a customer and track the journey." and a button "Open Customer Registration". Below this is a section for the "YOUR UNIQUE REFERRAL LINK" showing the URL `http://localhost:5173/ref-REF001` and a "Copy Link" button. A summary section displays four metrics: Clicks (1), Registrations (1), Purchases (1), and Commission (₹1,000). An "ACTIVITY" section titled "Your referral activity" contains a table with one row of data.

REFERRER	CUSTOMER	SERVICE	PURCHASE	COMMISSION	STATUS
—	Demo Customer	Skillwise Demo Service	₹10,000	₹1,000	Purchased

A — Owner/Admin dashboard showing complete activity



5. Current Demo Architecture

The proof-of-concept was deliberately separated from the live ADGA application so the business flow could be tested safely and explained to the client before production integration.

Layer	Demo implementation	Purpose
Frontend	React + Vite	Login, referral link, customer registration, service purchase, referrer/admin dashboards
Backend	Node.js + Express	Authentication, referral attribution, purchase handling, commission calculation and reporting APIs
Database	MongoDB Atlas	Stores users, referrals, services and purchase/referral activity
Authentication	JWT + bcrypt	Secure login and role-based access for Admin, Referrer and Customer
Commission	Configurable percentage	Current demo uses 10%; future rate changes can be handled through configuration

6. How This Should Be Implemented in the Real ADGA Website

The standalone demo proves the business logic. The production implementation should integrate the same concept into ADGA's existing user, course/service, authentication and payment systems rather than maintaining a separate application.

- 1. Referral creation: provide each approved referrer/partner with a unique referral code and shareable URL.
- 2. Referral attribution: when a visitor lands through the URL, store the referral attribution using a secure cookie/session or equivalent server-side mechanism.
- 3. Customer registration: attach the referral relationship to the newly created ADGA customer account.
- 4. Purchase attribution: when the customer completes a real ADGA course/service purchase, connect the successful order/payment to the referring partner.
- 5. Commission calculation: calculate commission only from a verified successful payment/order, using a configurable commission rate.
- 6. Admin reporting: add an admin referral dashboard showing referrer, customer, product/course, order amount, commission amount, status and timestamps.
- 7. Referrer dashboard: provide approved referrers with their link, clicks/leads, registered customers, purchases and commission status.

- 8. Payment integration: connect commission generation to the real payment confirmation/webhook rather than a demo purchase button.
- 9. Commission lifecycle: support statuses such as Pending → Approved → Paid, with appropriate admin controls and audit records.
- 10. Security: protect referral APIs, validate ownership/roles, prevent referral manipulation, duplicate attribution and unauthorized commission changes.
- 11. Testing: test referral link attribution, registration, successful/failed payments, duplicate clicks, refunds/cancellations and commission edge cases before production release.

Recommended production flow

Partner/Referrer → unique ADGA referral URL → Visitor → registration/lead → real ADGA course/service → successful payment webhook → commission calculation → Admin approval/payment → reporting for A and B.

7. Final Outcome

The problem was solved by turning the client's referral idea into a measurable end-to-end workflow instead of treating the referral link as a simple URL. The standalone demo confirmed that a referral can be attributed to B, C can register and purchase, the system can automatically calculate the configured commission, B can see the resulting commission, and A can see the complete referral activity. The next step for the real ADGA implementation is to integrate this proven flow with ADGA's existing authentication, customer/course data, orders and real payment confirmation system, while retaining configurable commission rules and proper admin controls.

Demo status: Core referral → registration → purchase → commission → admin tracking flow successfully tested.